

Adaptive Rotation Strategy with QQQ Rolling Selection Enhancement

1. Introduction

This project studies the Adaptive Rotation Strategy in the FinRL-Trading pipeline and extends it using my Assignment 1 QQQ rolling selection logic. The original strategy is a multi-layer rotation framework that dynamically allocates capital across different asset groups depending on market conditions, group strength, asset-level ranking, and risk controls.

The goal of this assignment is not only to reproduce the original Adaptive Rotation Strategy, but also to test whether a modified stock-selection layer can improve performance. My modification adds a rolling momentum-based filter inspired by QQQ selection logic. I then compare the original strategy, my modified strategy, SPY, and QQQ over the same backtest period.

The backtest period is:

2023-01-01 to 2024-12-31

2. Understanding the Original Adaptive Rotation Strategy

The Adaptive Rotation Strategy is built as a four-layer system:

1. Regime Layer
2. Group Selection Layer
3. Intra-Group Ranking Layer
4. Risk Control Layer

Each layer has a different role in the portfolio decision process.

2.1 Regime Layer

The regime layer identifies the current market environment. The strategy classifies the market into states such as:

- risk_on
- neutral
- fast_risk_off

The goal is to avoid blindly taking risk when market conditions deteriorate. For example, in a risk-on regime, the strategy can stay mostly invested. In a fast-risk-off regime, the strategy reduces exposure and raises cash.

In my backtest, the regime distribution was mostly risk-on:

fast_risk_off : 10 (9.8%)
neutral : 5 (4.9%)
risk_on : 87 (85.3%)

This means the test period was largely favorable for growth and momentum assets, especially technology stocks.

2.2 Group Selection Layer

The strategy divides assets into several groups, such as Growth Tech, Real Assets, Defensive assets, and Default market ETFs. Instead of always holding the same assets, the strategy calculates the strength of each group and chooses the strongest groups.

2.3 Intra-Group Ranking Layer

After the strategy chooses the strongest group, it ranks assets inside that group. The original strategy uses residual momentum.

The strategy prefers assets with strong positive residual momentum because those assets are outperforming for reasons beyond general market movement.

2.4 Risk Control Layer

The strategy also includes risk controls such as:

- fast risk-off triggers
- stop-loss exits
- cash allocation
- cooldown periods after stop-loss events

These rules help reduce exposure during stress periods. In the backtest, the strategy triggered multiple fast-risk-off adjustments and stop-loss exits. This confirms that the strategy was not simply buying and holding assets; it actively monitored risk during the test period.

3. My Modification: QQQ Rolling Selection Enhancement

For the modified version, I integrated my Assignment 1 QQQ rolling selection logic into the stock-selection process.

The original strategy ranks assets using residual momentum after group selection. My modification adds an additional filter before the residual momentum ranking step.

3.1 QQQ Rolling Logic

The rolling logic uses intermediate-term and long-term momentum. I calculate:

- 6-month momentum
- 12-month momentum
- skip the most recent period to reduce short-term reversal noise

The basic score is:

$$\text{Raw Momentum Score} = 0.5 \times 6\text{M Momentum} + 0.5 \times 12\text{M Momentum}$$

This follows the idea that assets with persistent medium- and long-term strength are more likely to continue outperforming.

3.2 Alpha Signal

The raw rolling momentum score acts as the alpha signal. The purpose is to identify assets with strong trend behavior before the original residual momentum step is applied.

Instead of passing every asset in a group into the residual momentum ranker, the modified strategy first narrows the candidate

pool to stronger rolling-momentum names.

3.3 Risk Control

I added a volatility-based risk adjustment. Instead of ranking only by raw momentum, the modified strategy penalizes high-volatility assets.

The risk-adjusted score is:

$$\text{Risk-Adjusted Momentum} = \text{Raw Momentum Score} / \text{Re}$$

This rewards assets that have strong momentum but avoids giving too much weight to unstable names. This is important because raw momentum can sometimes select assets that have high returns only because they are extremely volatile.

3.4 Applying the Filter Across Asset Groups

In the final version, I extended this enhanced rolling filter beyond only the Growth Tech sleeve. The filter is applied to active asset groups before the original residual momentum ranking step.

This means the final modified strategy still keeps the original architecture, but improves the candidate-selection process inside the strategy.

4. Original vs Modified Strategy Comparison

Performance Analysis			
Metric	Strategy	SPY	QQQ
Cumulative Return	2.08x	1.53x	1.94x
Annualized Return	44.83%	24.17%	40.04%
Annualized Volatility	20.31%	12.94%	18.27%
Sharpe Ratio	2.21	1.87	2.19
Max Drawdown	-13.24%	-10.12%	-9.95%
Calmar Ratio	3.39		
Win Rate	56.44%		

Figure 1. Statistics of QQQ Strategy

Performance Analysis			
Metric	Strategy	SPY	QQQ
Cumulative Return	1.98x	1.53x	1.94x
Annualized Return	41.18%	24.17%	40.04%
Annualized Volatility	23.86%	12.94%	18.27%
Sharpe Ratio	1.73	1.87	2.19
Max Drawdown	-15.93%	-10.12%	-9.95%
Calmar Ratio	2.59		
Win Rate	58.42%		

Figure 2. Statistics of Original Strategy

The modified strategy improved cumulative return, annualized return, volatility, Sharpe ratio, max drawdown, and Calmar ratio. The only metric that declined was win rate.

This is not necessarily a problem. Win rate only measures how often the strategy has positive periods. It does not measure how large the wins are or how severe the losses are. A strategy can have a lower win rate but still perform better if its average wins are larger or its losses are smaller.

That appears to be the case here. The modified strategy wins slightly less often, but it has better risk-adjusted performance and lower drawdown.

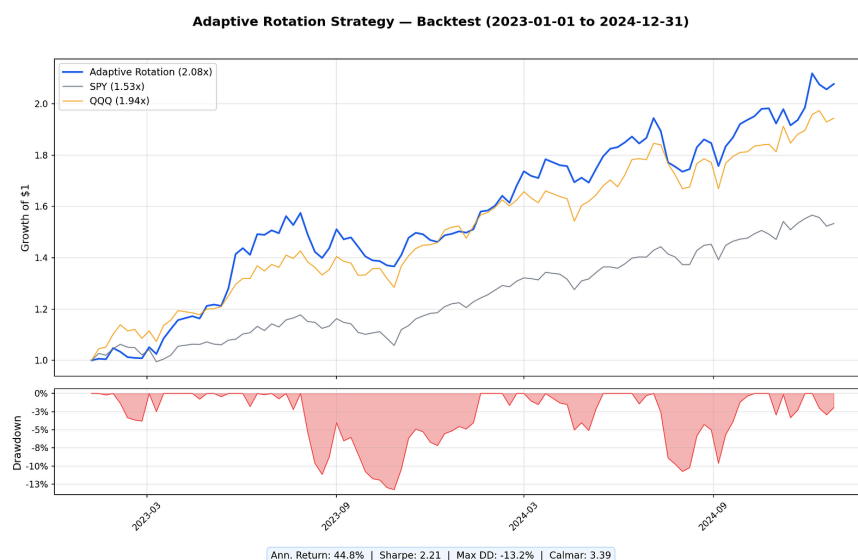


Figure 3. Performance of QQQ Strategy

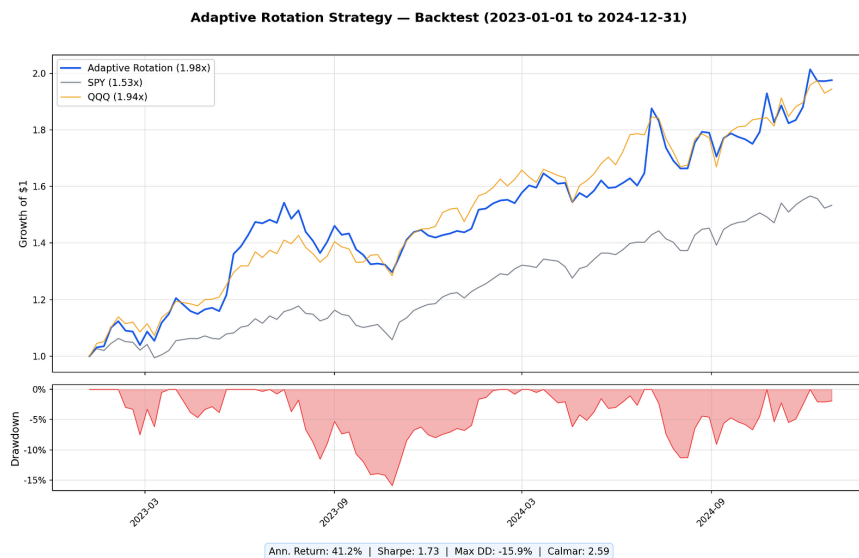


Figure 4. Performance of Original Strategy

5. Conclusion

This project reproduced the original Adaptive Rotation Strategy and enhanced it using QQQ rolling selection logic from Assignment 1.

The original strategy already performed well, achieving a cumulative return of 1.98x and an annualized return of 41.18%. However, it had relatively high volatility and drawdown.

The modified strategy improved the result. It achieved a cumulative return of 2.08x, annualized return of 44.83%, Sharpe ratio of 2.21, and max drawdown of -13.24%. Compared with the original version, the modified strategy produced higher return, lower volatility, better Sharpe ratio, better Calmar ratio, and smaller drawdown.

The only weaker metric was win rate, which declined from 58.42% to 56.44%. However, this is acceptable because the modified strategy improved the overall payoff profile. It did not win more often, but it produced better risk-adjusted results.

Overall, the experiment shows that adding a rolling momentum and risk-adjusted alpha filter can improve the Adaptive Rotation Strategy. The strongest evidence is the improvement in Sharpe ratio and max drawdown. Future work could test the strategy across longer time periods, add more robust regime detection, or compare different rolling momentum windows.